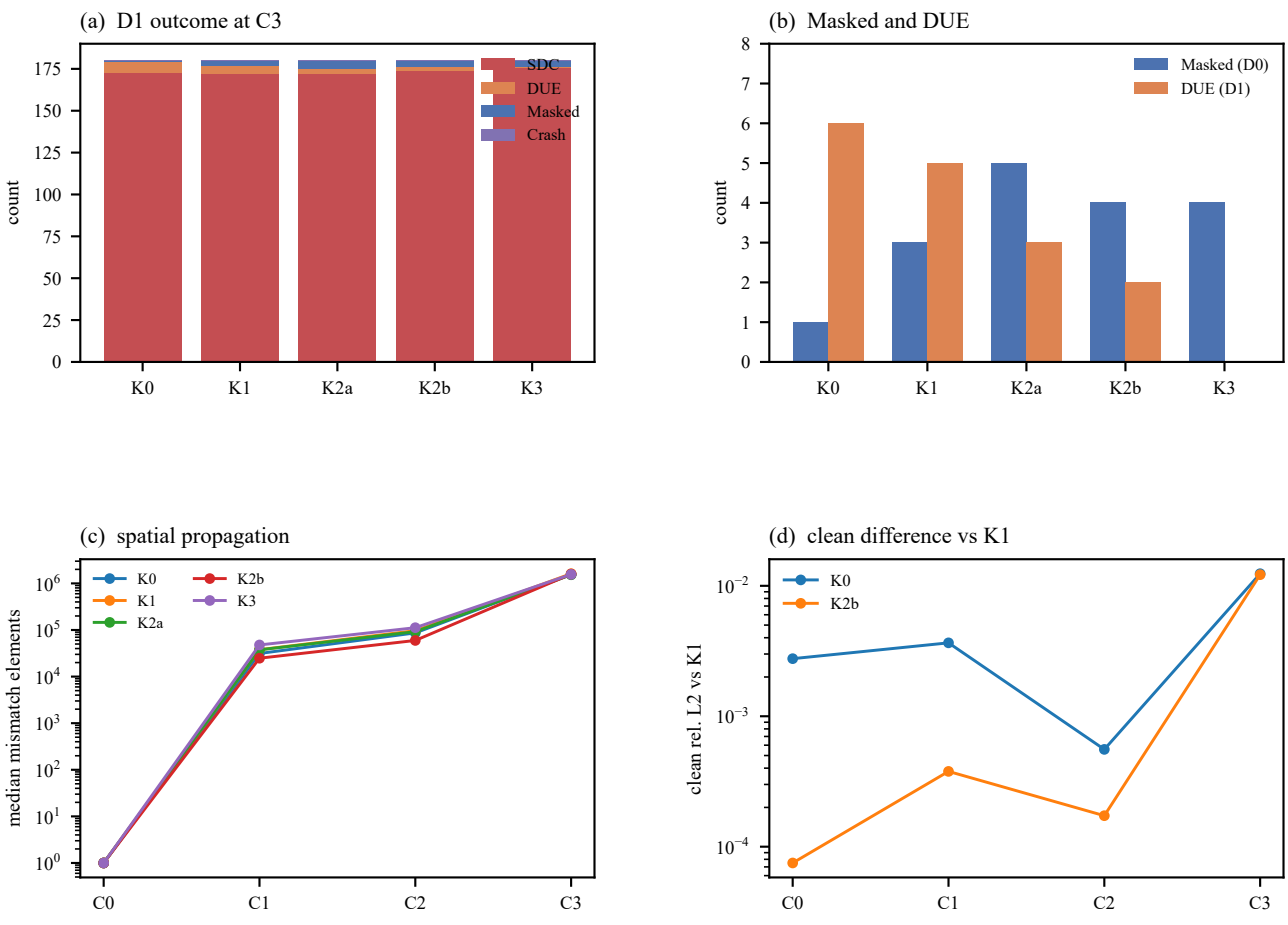
qkv_v kernel implementation comparison
K0 cuBLAS vs K1/K2a/K2b/K3 CUTLASS; 60 sites/bit; bits 28–30; C0→C3
(a) D1 outcome at C3
count
SDC
DUE
Masked
Crash
K0
K1
K2a
K2b
K3
(b) Masked and DUE
count
Masked (D0)
DUE (D1)
(c) spatial propagation
median mismatch elements
(d) clean difference vs K1
clean rel. L2 vs K1
C0
C1
C2
C3
K1/K2a/K3 clean outputs are bitwise identical; K2b clean differs. K0 uses cuBLAS + bias epilogue.